

AMIT MANDAL

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Education

National Institute of Technology Durgapur

Aug 2024 - Present

M.Tech. in Computer Science and Engineering · CGPA: 8.16

Cooch Behar Government Engineering College

Nov 2020 - Jul 2024

B.Tech. in Computer Science and Engineering · CGPA: 8.89

Akrurmoni Coronation Institution

May 2017 - Jun 2019

HS (12th) in Science · Percentage : 90%

Experience

Deloitte

May 2025 - July 2025

Data Analytics Job Simulation/Virtual Internship. [🔗](#)

- Performed data analysis on structured datasets to identify anomalies, trends, and potential fraud scenarios in digital forensic investigations.
- Generated insights to support risk assessment and data-driven decision-making in simulated client environments.

Accenture

June 2023 - Aug 2023

Developer Virtual Experience. [🔗](#)

- Defined technical requirements and enhanced system architecture, enabling cloud migration and implementing IAM policies for secure backend systems.
- Performed debugging, unit testing, and User Acceptance Testing (UAT) to ensure system reliability, while applying secure SDLC practices and conducting privacy assessments.

Projects

Deep Packet Inspection (DPI) Engine [🔗](#)

- Engineered a high-performance Deep Packet Inspection (DPI) engine in C++ with multi-threaded packet processing for efficient network traffic analysis and filtering.
- Built a protocol-aware packet parsing pipeline (Ethernet, IP, TCP/UDP) with deep packet inspection and SNI extraction, optimized via multithreading and hash-based load balancing for efficient traffic processing (YouTube, Facebook).
- Designed a scalable rule-based system to control network traffic based on application, domain, and IP.

Fuel Route Optimizer – Full Stack Route & Fuel Optimization Platform [🔗](#)

- Developed a full-stack fuel route optimization platform using Django REST Framework and React for long-distance trip planning across the USA.
- Integrated OpenRouteService API and geographic filtering to find nearby fuel stations and generate optimized routes in real time.
- Built fuel cost optimization and interactive map visualization using React, Tailwind CSS, Pandas & Geopy.

Water Quality Index Prediction [🔗](#)

- Developed a machine learning model to predict Water Quality Index (WQI) using CPCB data, applying feature engineering, model tuning, and algorithms like Random Forest and SVM.
- Performed data analysis and visualization using Pandas, NumPy, scikit-learn, and Seaborn to generate actionable insights.

Skills

Languages: C++, C, Python, JavaScript, SQL, HTML, CSS

Databases: MySQL, MongoDB

Frameworks & Libraries: React.js, Tailwind, FastAPI, Pandas, NumPy, Streamlit, Hugging Face, LangChain

Developer Tools & Platforms: Node.js, Docker, Postman, Git, GitHub, Linux, VSCode, Jupyter, Kaggle, Wireshark

Coursework: DSA · OS · OOPs · CN · DBMS

Soft Skills: Analytical Thinking · Problem Solving · Communication · Adaptability · Programming

Achievements & Interests

- Solved 150+ **LeetCode** problems, strengthening data structures, algorithms, and problem-solving skills. [🔗](#)
- Qualified **GATE** in 2024 & **WebJEE** in 2020.
- Selected to represent Malda district as a **Wicketkeeper-Batsman** at the inter-district level Cricket Tournament.
- Passionate about exploring new cultures, connecting with people & experiencing new landscapes.
- Deep interest in Cinematography, blending creative vision with technical skills.